Muhammad Ausaf Ayaz

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Portfolio: ausafayaz.github.io | GitHub: github.com/ausafayaz

Final-year Computer Science (Data Science) student seeking a 12-week internship in Karachi, Dec 2026 – Feb 2027

Education

Monash University Malaysia — Selangor, Malaysia
Bachelor of Computer Science (Data Science) — *Final Year – Currently Enrolled*

- **Computing coursework:** Algorithms and Data Structures, Databases, Programming Paradigms, Programming Fundamentals in Java, Artificial Intelligence
- **Data & maths coursework:** Introduction to Data Science, Modelling for Data Analysis, Data Visualisation, Foundations of Data Analysis, Discrete Mathematics, Continuous Mathematics

Karachi Grammar School — Karachi, Pakistan
A Levels — *2021 – 2023*

- A* Mathematics; A Physics; A English General Paper; B Computer Science; B Chemistry

Saint Michael's Convent School — Karachi, Pakistan
O Levels — *2019 – 2021*

- 8 A* and 1 A

Projects

More projects: github.com/ausafayaz

Young Malaysia Under Pressure | *Vega-Lite, JavaScript, HTML/CSS* — Live site

- Built an interactive data story of **12** Vega-Lite charts and maps from **4** official datasets (data.gov.my), covering youth unemployment, CPI and household income across **15** states and territories
- Designed a composite "pressure index" that ranks states on **3** normalised (0–100) indicators: unemployment, CPI change and spending-to-income ratio

APAC Superstore Profitability Dashboard | *R, Shiny, tidyverse, plotly* — GitHub

- Built an R Shiny dashboard with **5** interactive charts and **5** filters, joining **4** years (2011–14) of retail orders with **3** World Bank indicators
- Analysed **$3.59M** in APAC sales at a **12.2%** margin; showed loss-making orders rise sharply with discount level and Furniture earns the weakest margins

Flappy Birb | *TypeScript, RxJS, Vite, Vitest* — Play it

- Wrote a browser game in a purely functional reactive style: every input and timer is a pure reducer folded into one immutable state stream
- Implemented physics, collisions, CSV-driven levels, power-ups and ghost replays at **60** fps, covered by **18** unit tests with CI deployment on every push

Experience

Reet Solutions — Jun 2022 – Jul 2022
IT & Digital Operations Intern — *Karachi, Pakistan*

- Supported the planning and execution of digital and social media campaigns for company and client accounts
- Scheduled and published posts across social platforms and helped keep website content up to date
- Tracked basic engagement metrics (reach, likes, follower growth) and helped compile results for the team
- Assisted with day-to-day IT and digital operations tasks, troubleshooting minor issues and organising digital assets

Technical Skills

Languages: Python, Java, R, SQL, TypeScript, JavaScript, HTML, CSS
Data & Databases: Relational database design, SQL, normalisation, data cleaning, visualisation, statistical analysis
Software Development: Object-oriented programming, debugging, unit testing, functional reactive programming, front-end development
Foundations: Algorithms & complexity analysis, statistical modelling (regression, hypothesis testing), discrete & continuous maths
Tools & Libraries: Pandas, NumPy, Matplotlib, R Shiny, tidyverse, plotly, Vega-Lite, RxJS, Vite, Vitest, Git/GitHub

Leadership & Community

Monash University Malaysia: Organiser and volunteer for student-led clubs and campus events
Karachi Grammar School: First Aid Society member; Peer Tutoring Society tutor
The Last Earthlings: Volunteer, beach-cleaning community service initiative